AEBN25 - DAYLIGHTING AND LIGHTING OF BUILDINGS

MAIN ASSIGNMENT

Study Of Daylighting Conditions and Electric Lighting In "Home Offices"

LUND UNIVERSITY

Group No. 12

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Abstract

This study evaluates daylight performance in three student home offices in Sweden using a combination of physical measurements and Climate Studio simulations. Surface reflectance, window transmittance, and daylight factor (DF) were assessed under overcast conditions to compare real-world data with simulated results.

Findings reveal differences between measured and simulated daylight factors, attributed to variations in material properties, window transmittance, and external obstructions. Room Three, with large southeast and northwest-facing windows, exhibited the highest daylight provision but also a risk of glare. Room One, with northeast-facing windows, achieved moderate daylight levels but required electric lighting for extended periods. Room Two, heavily obstructed and equipped with low-transmittance glazing, suffered from inadequate daylight, necessitating artificial lighting throughout the day. None of the rooms met European daylighting standards (EN 17037).

These results underscore the importance of accurate material data, appropriate glazing selection, and strategic window placement in daylight optimization. The study highlights the need for enhanced daylight strategies to improve indoor comfort while minimizing energy use, particularly in Nordic climates with prolonged overcast conditions.

Keywords: Daylighting, Daylight factor (DF), Climate Studio simulation, Glare assessment, Surface reflectance, Window transmittance.

1 INTRODUCTION

Research indicates that effective daylighting offers numerous advantages, including enhanced health through circadian rhythm regulation, improved mood and well-being, greater productivity, better visual clarity, and higher energy efficiency. These outcomes are shaped by factors such as sunlight access, visual and thermal comfort, and energy efficiency, which form the foundation for daylighting design strategies (Dubois et al., 2019).

This project focused on evaluating daylighting conditions in students' homes by analyzing how natural and artificial light interact in spaces where they primarily work or study. Using Rhino 08 /ClimateStudio 0.2 (CS), simulations modelled each student's room or apartment, complemented by physical measurements of one real-world space. The goal is to compare simulated results with actual data, identifying discrepancies and exploring their causes.

To ensure realism in simulations, detailed measurements of one room were incorporated. This approach aimed at aligning virtual models with real-world conditions, improving the accuracy of CS simulations and providing insights into how materiality and spatial design influence daylight performance.

CONTRIBUTIONS

Each group member contributed to the project by modelling their home office. Material properties, simulation inputs, and results were collaboratively discussed during group meetings. Room measurements were conducted using instruments, with the attendance of all group members. All members also participated in writing the report.

2 METHODOLOGY

The study evaluated daylight performance across multiple home office environments in Sweden, leveraging group members' workspaces as case studies. Each space was analyzed, with surface reflectance, material properties, and spatial geometry documented to capture real-world interactions between light and built environments. Measurements were conducted under overcast sky conditions to align with diffuse daylight standards for one of the rooms, while simulations incorporated site-specific architectural features and material optical characteristics.

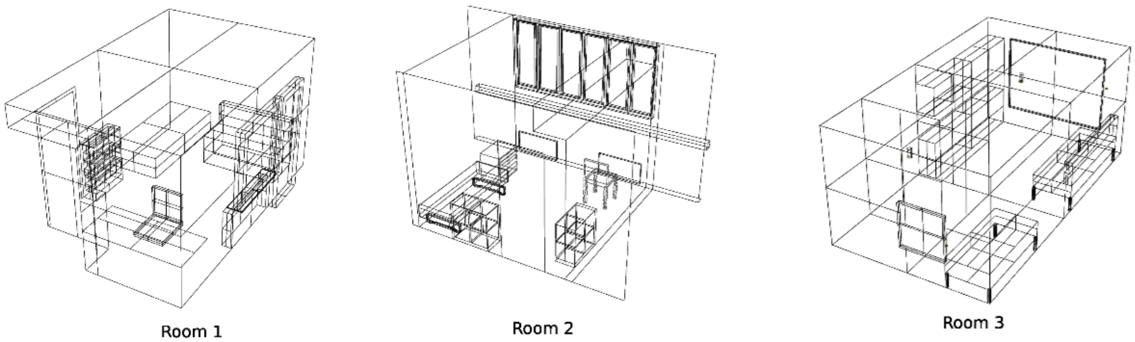

Figure 2.1 - schematic figure of selected rooms

2.1 SELECTED ROOMS

Simulations were performed for three rectangular home office spaces. While all rooms were modelled, Room 2 in Lund underwent additional in-depth analysis, including surface optical properties and daylight factor measurement to enhance data accuracy.

- **Room 1**: Located in Lund, this 12 m² ground-floor room is part of a two-story residential building. It features two northeast-facing windows running along the room’s length, providing daylight to the room.
- **Room 2**: Situated on the second floor of a five-story building in Lund, this 11.2 m² room includes a prominent east-facing window with mullions.
- **Room 3**: Occupying 28 m² on the fifth floor of an eight-story building in Malmö, this room integrates dual-aspect windows: a southeast-facing opening and a smaller northwest-facing window.

Table 1 - Geometry property of the selected rooms

Parameters	Room 1	Room 2	Room 3
Area (m^2)	12	11.2	28
Tot. Window size (m^2)	1.86	4.87	6,61
Orientations of windows	North-East	East	South-East & North-West

2.2 MEASUREMENT TOOLS

Some professional tools were used to measure the illuminance, glass transmissivity and surface reflectance. Two lux meters (Hagner model: EC1) were used to measure illuminance and glass transmissivity. For surface reflectance measurement a luminance meter (Hagner S4 photometer) and a reference plate with a white matt surface and known reflectance value (95.3 %) were used (Figure 2.2).

Figure 2.2 - Lux meter (left), photometer (middle) and reference plate (right) used for measurement

2.3 MEASUREMENTS

Room 2 was selected for full-scale measurements due to the accessibility for all group members.

2.3.1 DAYLIGHT FACTOR (DF) MEASUREMENTS

To measure the daylight factor, illuminance measurement was done for a grid of points in the room. A 0.5 m × 0.5 m measurement grid covered the floor with 0.45 m distance from all walls. With is method a grid contains 36 points were made in the room. The measurements were conducted on an overcast day (Figure 2.3). Horizontal illuminance at desk height (0.85 m) was recorded using two calibrated Hagner EC1 lux meters simultaneously for indoor and outdoor at the same time. For the outdoor measurement an open area without any shadowing effect of buildings or trees was selected. The process was repeated three times for each point.

Figure 2.3 - Overcast sky in measurement day

Daylight Factor (DF%) calculated using Equation 1. Daylight factor for each point considered as the average value of the three trials.

$$DF = \frac{\text{Indoor Illuminance}}{Outdoor\ Illuminance} x\ 100 \qquad \textit{Equation 1}$$

2.3.2 SURFACE REFLECTANCE (P) MEASUREMENTS

To measure the reflectance of opaque surfaces the photometer was held perpendicular to the surface, and the luminance was recorded. The process was then repeated with the reference plate with reflectance of 95.3%, and the entire procedure was performed three times for each surface. Since the reference plate had a known reflectance value, the reflectance of each surface was calculated based on the proportion of the measured luminance values. The final reflectance value for each surface was the average of the three trials. Surface reflectance measurements were taken for walls, door, floor, ceiling, desk, and window frame.

$$\rho_{SURFACE} = \frac{(L_{SURFACE}) \cdot (\rho_{PLATE})}{L_{PLATE}} \qquad \textit{Equation 2}$$

Where $\rho_{SURFACE}$ is reflectance value of the surface, $L_{SURFACE}$ is the luminance measured on the surface, ρ_{PLATE} is the reflectance value of the plate, and L_{PLATE} is the luminance measured on the plate.

2.3.3 WINDOW TRANSMITTANCE MEASUREMENTS

Two lux meters used to measure the window transmittance by measuring the illuminance immediately in front and behind the glass without shading it. The process repeated several times to get a reliable value. transmittance (Tn) was determined by calculating the ratio of indoor to outdoor illuminance (Equation 3).

$$\mathrm{Tn} = \frac{\text{Indoor Illuminance}}{Outdoor\ Illuminance} x\ 100 \qquad \textit{Equation 3}$$

The resulting value from above measurement and calculation is glass transmittance. To use this value as input in simulation it should be converted to transmissivity using Equation 4.

$$tn = \frac{\sqrt{0.8402528435 + 0.0072522239\,.\,\mathrm{Tn}\,.\,\mathrm{Tn}} - 0.9166530661}{0.0036261119 \cdot \mathrm{Tn}} \qquad \textit{Equation 4}$$

2.4 MODELLING AND SIMULATION

2.4.1 MODELLING APPROCH

A 3D model of all three rooms was created in Rhino 8 using measured dimensions, incorporating precise spatial parameters, material optical properties, and window transmission characteristics. The geometry was prepared with closed planar surfaces to ensure accurate light simulations. The modeling prioritized geometric accuracy for surfaces and fixed elements critical to daylight performance, including detailed wall assemblies, simplified furniture to capture light-blocking or reflecting effects, and adjacent buildings to simulate shading and reflected light interactions. Non-essential decorative details were omitted for computational efficiency, while light-affecting elements such as window frames, mullions, and furniture were retained to preserve realistic shading and light distribution.

2.4.2 MATERIAL ASSIGNMENT

The CS material workflow typically involves selecting predefined materials (e.g., Matte White Plaster, Clear Glass, Dark Wood Floor) from its built-in library or developing custom materials by specifying reflectance (diffuse), transmittance (for glazing), and specularity parameters within the Material Editor to achieve realistic light behavior in simulations. For Rooms One and Three, materials were selected from the CS library using educated estimates for surface reflectance values and glass transmissivity. Room Two, however, utilized custom materials generated via theRadGen Tool ("Radiance Material Generator · Streamlit," n.d.), where reflectance values were determined through physical measurements and colors were sampled using the mobile Android app *Color Picker AR*. Glazing material was matched to the closest equivalent in the CS library, prioritizing identical properties, pane configuration, and color. In cases where measured values were deemed unreliable, standard materials from the CS library were applied as substitutes.

2.4.3 POINT-IN-TIME LUMINAIRES

The CS's workflow calculates illuminance distributions for electric lighting and daylight at specific moments in time. Luminaires are modelled by default or importing manufacturers-supplied IES files, with defined properties such as luminous flux and power. Fixtures are positioned per architectural plans, accounting for ceiling height, orientation, and lighting type (direct/indirect).

2.4.4 DAYLIGHT FACTOR (DF)

The Daylight Factor quantifies the illuminance inside a room (typically on a horizontal plane) as a percentage of the total light available from an unobstructed overcast sky (Lewis, 2017). The DF analysis assesses static daylight performance by preparing the models with sensor grids in each room. A 0.5 m margin from the walls was excluded, and spacing points were placed 0.5 m apart at a height of 0.85 m. Simulation was done to calculate DF values for each grid point, and DF_{mean}, DF_{median}, and Uniformity ratio for rooms.

2.4.5 CLIMATE-BASED DAYLIGHT

The Climate-Based Daylight Analysis evaluates annual daylight performance using real-world weather data. The Copenhagen Kastrup EPW file from the CS Library was used for Room One and Room Three, while the weather file for Lund from the same library was used for Room Two to achieve more realistic results. Since this simulation is location-dependent, the real orientation of the rooms was applied to the files. The simulation was limited to occupied hours, from 8 AM to 6 PM every day. A sensor grid like the DF analysis (0.85 m height, 0.5 m spacing/margin) was used. Simulations were conducted for target illuminance thresholds (100, 300, 500, and 750 lux) to assess compliance with European Standard EN 17037. Results include metrics such as Spatial Daylight Autonomy (sDA) and Annual Sunlight Exposure (ASE), and Daylight Autonomy (DA).

2.4.6 DAYLIGHT GLARE (DG)

Glare refers to a condition when the brightness contrast in the visual field is excessively high, making it difficult for the eye to adjust (Dubois et al., 2019). The Daylight Glare Probability (DGP) estimates the probability that an observer, based on their view position and orientation, will experience discomfort glare ("Annual Glare — ClimateStudio latest documentation," n.d.). To assess glare in CS, an annual glare simulation was run for each room. A critical position in the room was selected by help of the plan view for annual glare simulation and sDG values. sDG represents the percentage of the views with disturbing glare more than 5% of the time. A key time point was chosen to represent moments when glare is most likely to occur. Using real-world weather data for the location, simulations are conducted to calculate DGP at eye level (approximately 1.2 m above the floor). Rendered images of luminance distribution were generated for the time point, and the results were analyzed to evaluate occupant visual comfort.

2.4.7 ENERGY USE FOR ELECTRIC LIGHTING

Daylight Autonomy (DA) defines the percentage of the occupied hours of the year when the minimum illuminance threshold at a point or a grid of points is met by daylight alone (Reinhart and Walkenhorst, 2001). Light Dependency (LD) represents how much space depends on artificial lighting. LD can be calculated by Equation 5 by knowing the DA value from simulation for 300lx target. Annual energy use for electric lighting calculated by multiplying the power of existing artificial lights in each room, by LD for occupied hours.

$$LD = 1 - DA \qquad \textit{Equation 5}$$

2.4.8 SUBJECTIVE EVALUATION

To make visual inspections of the space and make a subjective judgement of daylight in the rooms, two questionnaires were used. The first one was Daylight survey which filled individually by the student who accommodate in each room, and the second one was Belupp (Küller, Wetterberg, 1993) for light experience, which filled by each group member and only for the room that was measured. The Belupp questionnaire had 18 items loading on five factors: hedonic tone, brightness, variation, color, and flicker. The result translated from score items into factors using a scoresheet.

3 RESULTS

3.1 MEASURED VALUES FOR ROOM TWO

Table 2 illustrates the reflectance values for diffusive surfaces in Room Two based on the measurement and calculation. As can be seen the value for ceiling is not acceptable. The resulting value for glass transmittance and transmissivity was 12.50 and 13.61, respectively. Based on the measurement and calculations the DF $_{mean}$ equals to 0.29 % and DF $_{median}$ equals to 0.24 %.

Table 2. Reflectance values of surfaces in Room Two

Object	$L_{SURFACE}$ (cd/m²)	L_{PLATE} (cd/m²)	$\rho_{SURFACE}$ (%)
White painted wall	19.33	14.33	70.65
Door	30.00	20.00	63.53
Ceiling	7.67	22.33	277.61 (error!)
Window frame	33.67	12.67	35.86
Floor	22.67	8.00	33.64
Windowsill	20.67	16.33	75.32
Desk	13.33	12.00	85.77

3.2 MEASURED AND SIMULATED DAYLIGHT PERFORMANCE OF ROOM TWO

Figure 3.1 – Photo of Room 2

Table 3 illustrates the results of daylight performance of Room Two based on simulation and measurement. Also, Figure 3.2 demonstrates DF values for grid points achieved by two methods. The values for the measurement process can be found in Table 01 1 in Appendix 01.

Table 3 - results for daylight performance in room Two based on simulation and measurement

Simulation			Measurement		
DF $_{mean}$ (%)	**DF $_{median}$ (%)**	**Uniformity ratio**	**DF $_{mean}$ (%)**	**DF $_{median}$ (%)**	**Uniformity ratio**
0.30	0.28	0.46	0.29	0.24	0.37

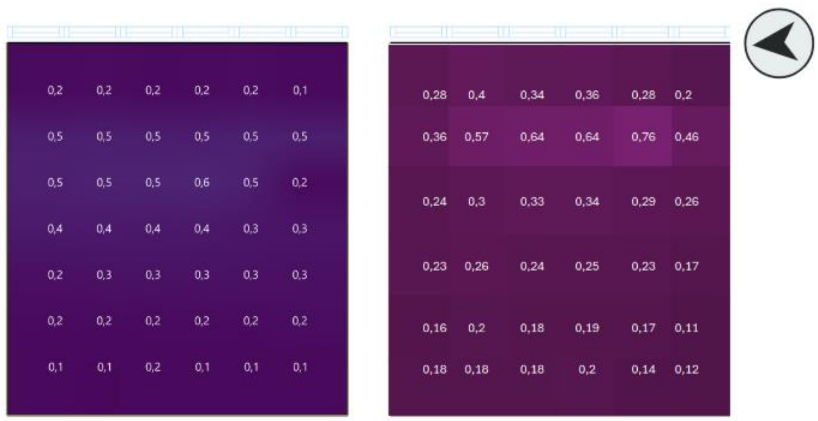

Figure 3.2 - values for DF in the grid of Room Two based on simulation (left) and measurement (right)

3.3 DAYLIGHT PERFORMANCE OF THE ROOMS

The following section presents the simulated daylight performance of the three rooms.

3.3.1 DAYLIGHT PROVISION

The results of the simulations are present when doing the simulation for DF under overcast sky in Table 4 and Figure 3.3 shows the DF values for each grid point in the rooms. The three different rooms simulate 3 different results regarding DF. Room Three has a higher exposure to daylight, this is shown by the yellow-colored area of the room in Figure 3.3.

Table 4 - Result of the simulation for DF under overcast sky

	DF $_{mean}$ (%)	**DF $_{median}$ (%)**	**Uniformity ratio**
Room 1	0.87	0.82	0.69
Room 2	0.30	0.27	0.46
Room 3	1.9	1.5	0.10

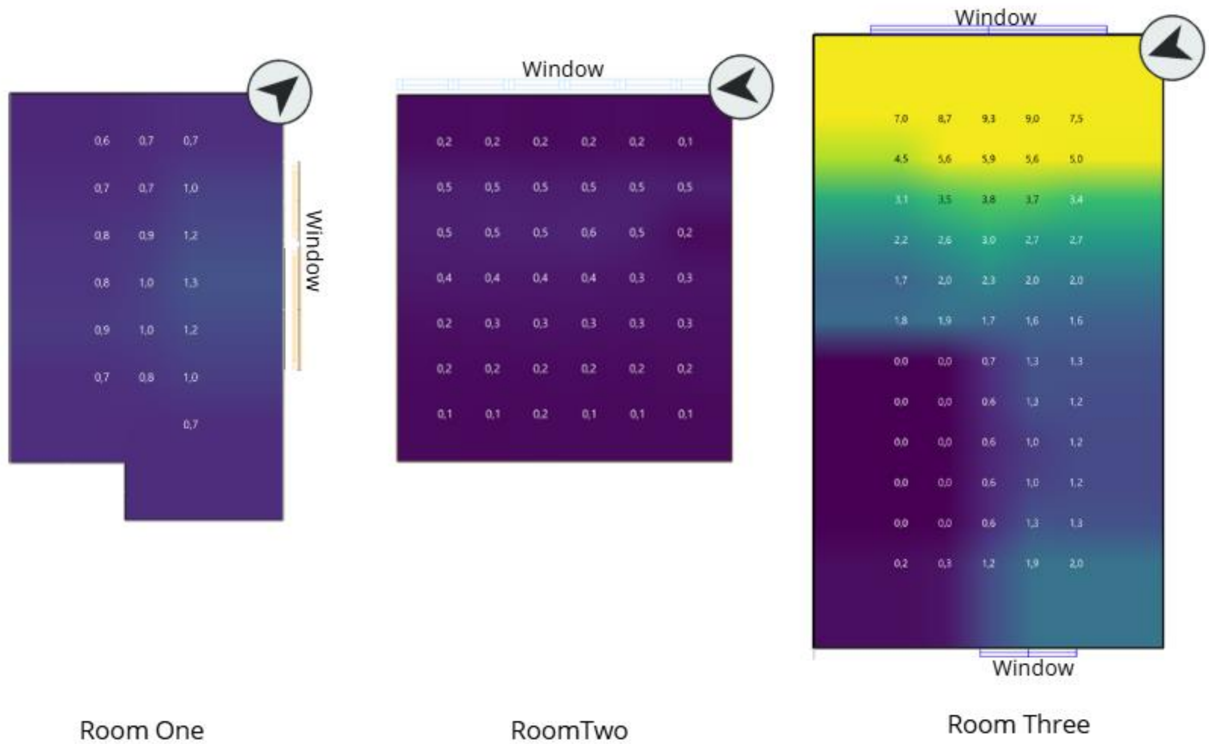

Figure 3.3 - Simulation result for DF values for each grid point in the rooms

Table 5 illustrates the results of the climate-based simulation for 100, 300, 500 and 750 lux.

Table 5 - Results for sDA in climate-based simulation at 100, 300, 500, 750 lx.

	sDA $_{100/50\%}$ (%)	**sDA $_{300/50\%}$ (%)**	**sDA $_{500/50\%}$ (%)**	**sDA $_{750/50\%}$ (%)**
Room One	94.70	0	0	0
Room Two	2.40	0	0	0
Room Three	70.0	48.0	24.0	10.0

Table 6 shows the simulation results for the Useful Daylight Illuminance (UDI) metric in rooms. This metric was initially designed to evaluate daylight levels at specific points or grids within a room (Dubois et al., 2019). CS 0.2 categorized illumination as Failing (UDI.f) (<100 lux), Supplemental (UDI.s) (100–200 lux), Autonomous (UDI.a) (300–3000 lux), or excessive (UDI.e) (>3000 lux) ("Custom Daylight Availability — ClimateStudio latest documentation," n.d.)

Table 6 - Simulation result For UDI in rooms

	UDI.f (%)	UDI.s (%)	UDI.a (%)	UDI.e (%)
Room One	37.00	45.37	17.63	0
Room Two	74.5	20.81	3.78	0.91
Room Three	39.87	19.1	40.23	0.79

3.3.2 DAYLIGHT GLARE

Based on the simulation results in CS, Room One, Two, and Three have sDG values of 0, 3.6%, and 22.8%, respectively. This means that occupants in Room One never experience disturbing glare, while in Room Three, 22.8% of the views are affected by disturbing glare for more than 5% of the time. Room Two falls in between, with 3.6% of views experiencing similar glare conditions. In CS 0.2, DGP is classified into four categories based on values ranging from 0 to 100% (Figure 2.2).

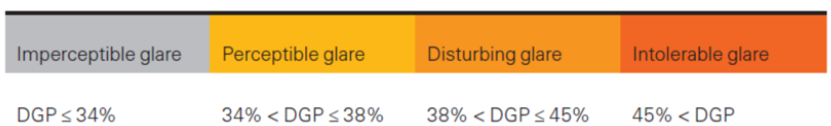

Figure 3.4 classified categories for DGP in CS 0.2

As can be seen from the above values there is the minimum risk of experiencing glare in Room One. There was only one sensor in front of the window with the risk of distributing glare during summertime in the morning. Figure 3.5 shows the situation on 20 Jun 8:30 AM when DGP is equal to 0.40.

Figure 3.5 - Glare assessment in Room One (20 Jun 8:30 AM), Room Two (1 Jul 12:30 PM), Room Three (19 Apr 8:30 AM).

Room Two experiences very little direct sunlight due to its location and the building's design, which limits such exposure. However, the analysis reveals that occupants may encounter minimal direct sunlight on a few days in mid-June and mid-July. Figure 3.5 shows the situation on 1 Jul 5:30 AM when DGP is equal to 0.73.

Room Three experiences disturbing glare that would make it uncomfortable for a person living there. The glare assessment in room three (19 Apr 8:30 AM) is experiencing a lot of direct sunlight. The simulation gave the result that the DGP is equal to 0.95 and there will be a lot of Intolerable Glare, which means it´s well above 45 % in DGP.

3.3.3 ENERGY USE FOR ELECTRIC LIGHTING

To calculate energy use for electric lighting, the powers of existing lights were read. Room One has three same lamps with 6.3 W, Room Two has one central drop light with 12 W and Room Three has two lamps with a power of 7 W, respectively two drop lamps with a power of 9 W.

Table 7 - 3.3.3 Energy use for electric lighting in rooms

	DA (at 300lx target) (%)	LD (%)	Annual occupied hours	Power of existing lights (W)	annual energy consumption for lighting (kWh/year)
Room One	17.63	82.37	3650	18.9	56.80
Room Two	4.09	95.91	3650	12	42.00
Room Three	41.03	58.97	3650	32	69.97

The results indicate that Room Three has relatively higher energy consumption for electric lighting compared to the other rooms, despite being the most daylit space. This can be attributed to its larger area. Additionally, occupants' experiences reveal that electric lighting sources are not used simultaneously in all rooms. For instance, in Room One, only the desk lamp is typically used during study hours, and the lighting quality in other parts of the room is not a concern at that time. This means that the energy used for lighting would be lower than the calculation.

3.3.4 SUBJECTIVE EVALUATION

Subjective evaluations of daylight quality in the three rooms provide valuable insights into occupants' perceptions and feelings beyond the quantitative standards and simulations. This evaluation reveals differences in occupant perceptions, emphasizing the importance of integrating user feedback (Table 8 - Summary of daylight survey questionnaire). In Room One, occupants experience slightly discomforting daylight conditions, with 65% of the space perceived as insufficiently lit, leading to dissatisfaction and reliance on electric lighting for 40% of the day. Room Two is perceived as having low daylight levels, with 100% of the space considered insufficiently lit and full reliance on electric lighting throughout the day. Conversely, Room Three is rated highly daylit by occupants, with only 7% reliance on electric lighting, while most areas experiencing sufficient daylight. These findings highlight the value of subjective assessments in identifying user-specific needs that may not be fully captured by simulations or standards alone.

Table 8 - Summary of daylight survey questionnaire

Subjective	**Room One**	**Room Two**	**Room Three**
Daylight Perception	Neither high nor low	Low	Extremely high
Daylight condition	Slightly discomfort	No discomfort	No discomfort
Occupants' preference for the room	Brighter	Brighter	As it is
Percentage of space with excessive, sufficient, and insufficient daylight	0, 35%, 65%	0, 0%, 100%	65%, 30%, 5%
Usage of electric light during the day	40%	100%	7%
General daylight condition	Unacceptable	Unacceptable	Acceptable

4 DISCUSSION

4.1 COMPARISON BETWEEN MEASURED AND SIMULATED VALUES

When we compared real-world measurements to simulations in Room Two, the daylight factors were close—like the DF_{mean} being 0.29% in reality versus 0.30% in the model. But the gap was in how evenly light spread across the room: real measurements showed a uniformity ratio of 0.37, while simulations predicted 0.46. Some of this mismatch comes from human or natural errors—like the light meters having a 3–5% margin of error, or the sky not staying perfectly overcast during measurements.

The window transmittance measurement led to some confusion, as the lux meter readings were very low, resulting in a calculated transmissivity of just 13% for the glass. This seemed unreasonable since the glass appeared clear. To verify the results, the measurements were repeated multiple times throughout the day, but the values remained within the same range. Based on these findings, the most similar glazing material from the CS library was selected for the simulation (Visible Light Transmittance (VLT): 15%, double-glazed, reflectance: 8%). Surprisingly, the DF results with this glazing closely matched the real-time measurements, as shown in Figure 3.2. Given this consistency with real-world conditions, this glazing material was used for the next simulation steps.

To further evaluate the importance of the glazing properties, additional simulations were conducted using different VLR values, while keeping the reflectance and panel properties constant. Table 9 illustrates how daylight provision can be improved by using higher transmittance glass. However, since overheating can also be an issue, advanced assemblies known as spectrally selective glazing, which feature coatings that allow twice as much visible light as solar heat, have become a common choice for balancing high light transmittance with low solar heat gain (Dubois et al., 2019).

Table 9 - Result for daylight provision in Room Two with different glazing

VLT of Glazing	DF_{mean} (%)	DF_{median} (%)	Uniformity ratio	$sDA_{300/50\%}$ (%)
15%	0.30	0.27	0.46	0
36%	0.78	0.74	0.44	0
60%	1.27	1.17	0.41	35.07

There were also some conditions of errors in measurement. For instance, ceiling's reflectivity somehow was measured at an impossible 277.61%, probably because the light meter was not angled right, or the surface was not diffusive.

4.2 COMPARISON BETWEEN ROOMS

The three rooms have some crucial differences that can affect the daylighting performance. Some differences are orientation, window size and depth, frames, location of the apartment, daylighting being blocked by nearby buildings, etc.

Room One is a rectangular space located on the ground floor, featuring a clear north-east-facing window. The window faces a one-story building at an acceptable distance, resulting in a sky exposure angle of 74°, as shown in Figure 4.1, allowing adequate daylight entry into the room. The ratio between the window height and the room depth is approximately 1.3, which supports good light distribution across space. Despite these design advantages, the DF_{mean} was measured at 0.7, placing the room in the acceptable category but indicating reliance on electric lighting for adequate illumination.

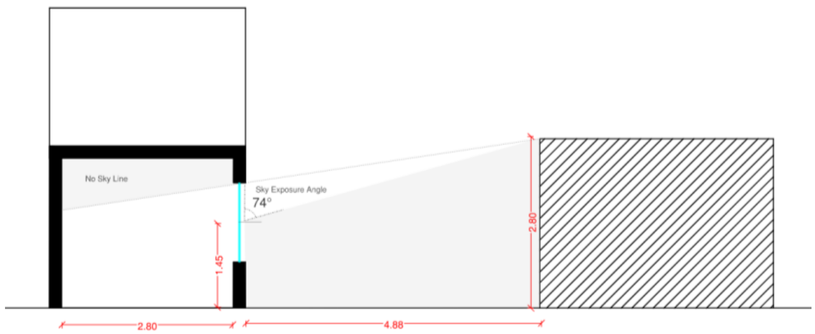

Figure 4.1 - Condition of Room One in neighbouring area

Occupant's feedback revealed that electric lighting is particularly necessary when working in areas farther from the window, such as the study desk positioned in the corner of the room.

Figure 4.2 - Daylight condition of Room One in a winter day at 13 PM in a partly cloudy day

To improve daylight quality and uniformity, potential solutions include increasing the window width or dividing it into two sections to enhance light penetration to all corners of the room. Extra simulation was done to assess the possibility of this theory. Result showed that by having a slightly bigger window area and dividing it into two window DF_{median} can improve from 0.82 to 1.1 (Figure 4.3

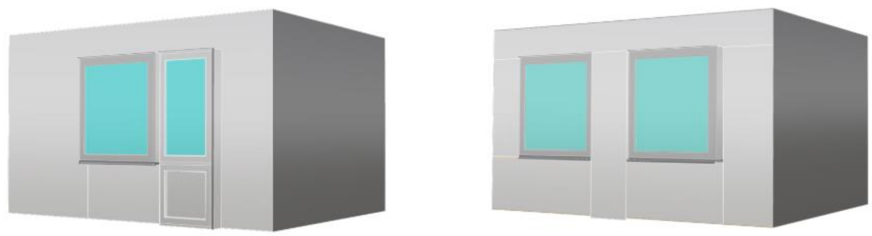

Figure 4.3 - Improvement suggested for room One for better daylight factor (right) compared to room situation (left)

Room Two, with a square plan and a window spanning the full width of the room, had significantly low DF values and insufficient daylight provision compared to the other rooms. The primary cause of this issue is the limited sky exposure angle, which is influenced by external obstructions, including neighboring buildings and a deep overhang above the window. As illustrated in the Figure 4.4, the room is located on the second floor, and the extended overhang combined with the nearby building results to a restricted sky exposure angle (22°), limits natural daylight penetration into the interior space. Additionally, when viewed in a broader context, the room is situated in a shaded area caused by another high-rise building directly opposite it, restricting the entry of natural daylight to the room. Another contributing factor to the poor daylight quality is the design of the window itself. Despite its wide dimensions, the window has low visible transmittance and starts at a height of 1.6 m above the floor, which reduces its effectiveness in distributing daylight across the room. The ratio between the depth of the room and the window is approximately 1.2, which under ideal conditions could support good daylight distribution. However, due to the restricted sky exposure, shading from adjacent structures, low window transmittance, and elevated window placement, the room's daylight performance is not acceptable.

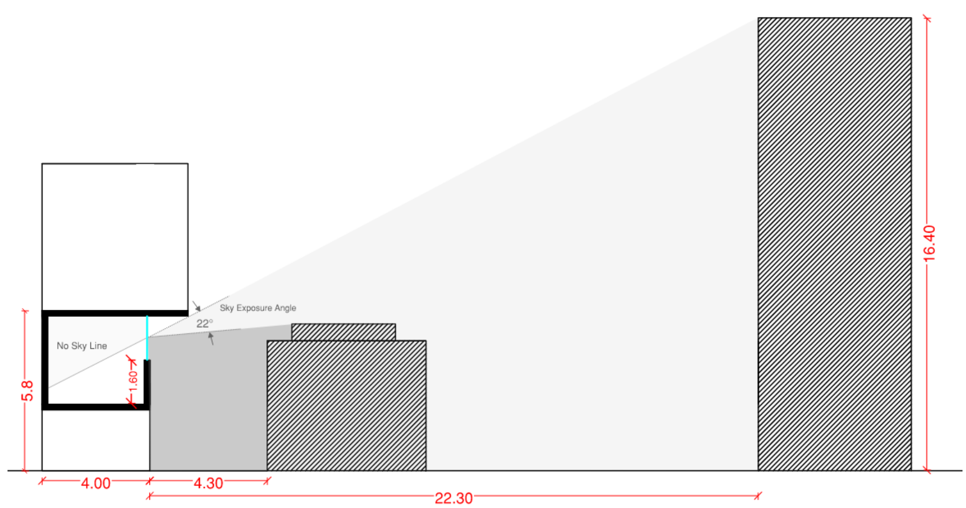

Figure 4.4 - Condition of Room Two in neighbouring area

Room Three, located on the fifth floor, features a rectangular plan with a large east-facing window made of clear glass. The context building has minimal interference with daylight access, resulting in a sky exposure angle of 76°, as illustrated in Figure 4.5. The ratio between the window height and the depth of the room is close to three, which is more than the recommended value, but the high sky exposure of the room can compensate for this and make it quite daylit. The east façade has a window-to-wall ratio of 50%, which allows ample daylight entry into the front part of the room but also increases the risk of glare due to direct sunlight exposure. Despite this, daylight provision to the rest of the room is very low, leading to poor uniformity in light distribution. The calculated uniformity ratio is 0.08, indicating that daylight is unevenly distributed across space. As shown in Table 5 $sDA_{100/50\%}$ and $sDA_{300/50\%}$ was 70% and 48%, respectively, The highest values among the three rooms.

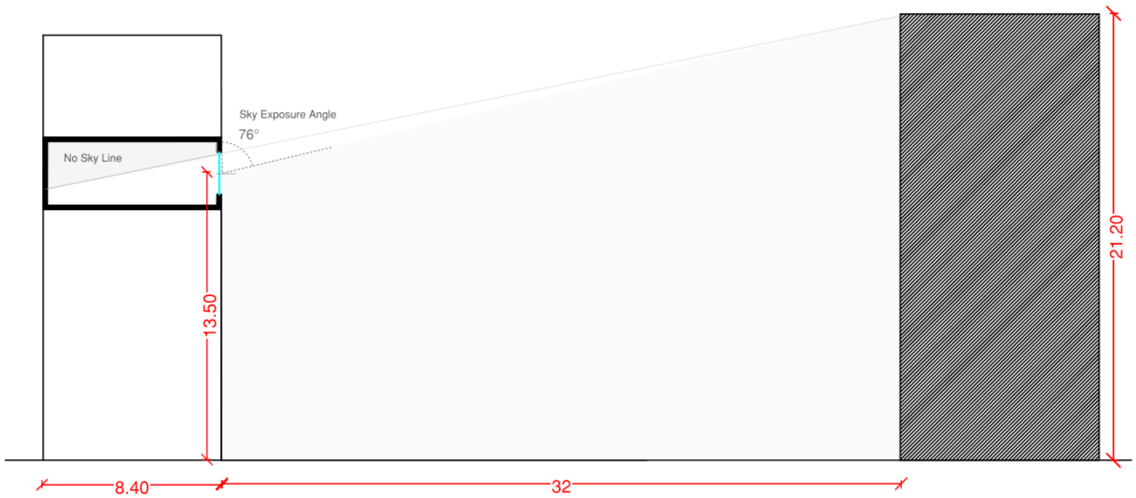

Figure 4.5 - Condition of Room Three in neighbouring area

4.3 ASSESSMENT OF THE ROOMS WITH EUROPEAN STANDARDS

Light is particularly significant in the Nordic countries due to its scarcity for much of the year and abundance during the summer solstice. Domination condition of overcast sky in this region make the DF value very important in indoor daylight conditions (Dubois et al., 2019). Table 10 illustrates the Interpretation for different values of the DF in the room (Tregenza and Wilson, 2013). Based on this categorization, Room One and Three can be considered close to acceptable balance between electric lighting and daylighting (DF_{median}: 0.7 – 1.5%) and Room Two has dark and gloomy impression. As can be seen in Figure 3.3 Room Three has higher amount of direct sunlight compared to Room One, but there are also several sensors faraway from the window with low value in DF and the small window at the back of the room cannot be helpful in that regard due to the small size.

Table 10 - Interpretation for the Daylight Factor in room (Tregenza and Wilson, 2013)

DF_{mean}	DFp or DF_{median}	Description
< 1%	< 0.7%	Dark, gloomy impression. Electric lighting hides daylight variations.
1–2%	0.7–1.5%	Acceptable balance between electric lighting and daylighting.
4–5%	3.0–3.8%	Experience of a totally daylit room.
> 5%	> 3.8%	May create thermal discomfort, noise, or glare.

For climate-based assessment European standard EN 17037:2018+A1:2021 ("Standard - Dagsljus i byggnader SS-EN 17037," n.d.) was used to evaluate if the rooms meet the recommended performance for daylight provision (Table 11). Based on this standard to achieve the minimum recommendation value the room requires to meet the two following criteria.

- **Target illuminance:** Minimum of 300 lux must be achieved over 50% of the floor area for no less than 50 % of the daylight hours.
- **Minimum illuminance:** Minimum of 100 lux must be achieved over 95 % of the floor area for no less than 50 % of the daylight hours.

Table 11 - Level of Recommendation for Vertical and Inclined Daylight Opening based on EN 17037:2018+A1:2021

Level of Recommendation for Vertical and Inclined Daylight Opening	Target Illuminance (E_T) (lx)	Fraction of Space for Target Level (F_{plane}) (%)	Minimum Target Illuminance (E_{TM}) (lx)	Fraction of Space for Minimum Target Level (F_{plane}) (%)	Fraction of Daylight Hours (F_{time}) (%)
Minimum	300	50%	100	95%	50%
Medium	500	50%	300	95%	50%
High	750	50%	500	95%	50%

Based on the simulation conducted for different target illuminance thresholds (100, 300, 500, and 750 lux) (Table 5). none of the rooms could meet the requirements. Even Room Three, which had the best condition among all, could not reach the Minimum of 100 lux over 95 % of the area. Figure 4.6 shows the values for *sDA* at 300lx target simulation. This shows that it is not easy to meet the recommended standards of daylight provision and how many parameters need to be done wisely in design stage to have the best possible daylight performance in their building.

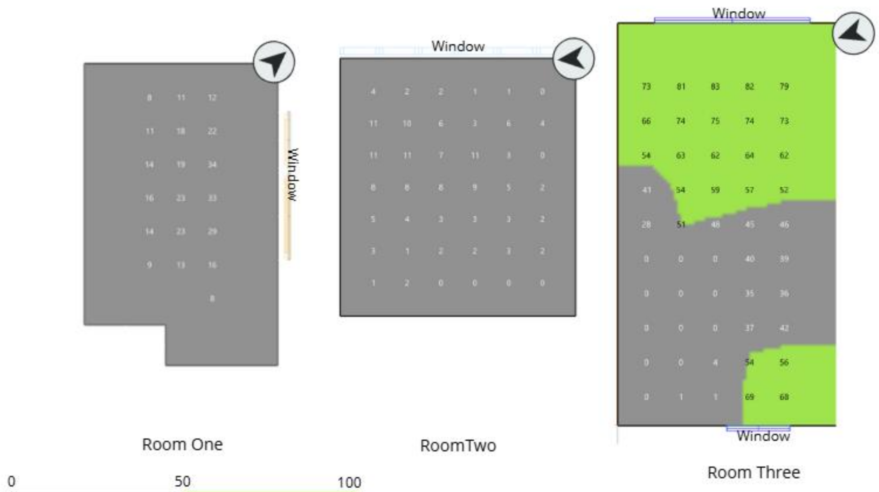

Figure 4.6 - values for sDA at 300 lx target simulation

5 CONCLUSIONS

This study showed that good daylight in buildings is not just about following rules or hitting technical targets, it is a balance of orientation, glazing properties, room size and shape. Also, external obstructions significantly impact daylight availability and visual comfort in rooms. The bigger windows do not always guarantee better light. Room Three had large windows facing two directions, which gave it balanced daylight throughout the day. But Room Two, despite having a big window, barely got any light because it was overshadowed by nearby buildings. This proves that strict rules about window size are not enough, orientation and context matter just as much. The findings reveal that while large windows can improve daylight access, they may also introduce glare issues, as seen in Room Three. Conversely, poor window placement and low transmittance glazing, as in Room Two, can severely limit daylight provision, increasing reliance on artificial lighting.s

A highlighted result of the study is that none of the rooms met the EN 17037 daylight recommendation, demonstrating the challenge of designing naturally well-lit spaces in overcast Nordic conditions. The discrepancies between measured and simulated values suggest that accurate input data, particularly for material properties and glazing performance, is crucial for reliable daylight modelling.

These insights emphasize the need for thoughtful daylighting strategies in early stages of architectural design, balancing daylight access, glare control, and energy efficiency. Future studies should explore alternative shading solutions, adaptive glazing technologies, and seasonal variations to refine daylighting strategies further.

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APPENDIX 01

Table 01 1 - total values for daylight factor measurement in Room two

	Indoor			Outdoor						
point No.	trial1	trial2	trial 3	trial1	trial 2	trial 3	DF1	DF2	DF3	$DF_{average}$
1	9	10	9	333	334	336	0,27	0,30	0,27	0,28
2	11	16	13	335	336	340	0,33	0,48	0,38	0,40
3	12	12	12	353	354	352	0,34	0,34	0,34	0,34
4	12	12	12	336	331	329	0,36	0,36	0,36	0,36
5	7	7	6	244	239	235	0,29	0,29	0,26	0,28
6	5	5	4	231	231	230	0,22	0,22	0,17	0,20
7	10	11	10	287	285	288	0,35	0,39	0,35	0,36
8	16	16	17	288	289	289	0,56	0,55	0,59	0,57
9	18	19	18	288	287	285	0,63	0,66	0,63	0,64
10	17	18	19	273	282	285	0,62	0,64	0,67	0,64
11	20	19	19	246	255	260	0,81	0,75	0,73	0,76
12	11	11	11	237	239	242	0,46	0,46	0,45	0,46
13	7	6	8	294	294	295	0,24	0,20	0,27	0,24
14	9	8	9	294	294	293	0,31	0,27	0,31	0,30
15	10	9	9	288	287	286	0,35	0,31	0,31	0,33
16	10	10	10	288	293	293	0,35	0,34	0,34	0,34
17	8	9	9	295	296	294	0,27	0,30	0,31	0,29
18	8	7	7	288	284	280	0,28	0,25	0,25	0,26
19	3	3	3	128	129	130	0,23	0,23	0,23	0,23
20	3	4	3	127	127	127	0,24	0,31	0,24	0,26
21	4	3	3	140	135	134	0,29	0,22	0,22	0,24
22	4	4	3	153	147	143	0,26	0,27	0,21	0,25
23	4	5	3	186	175	162	0,22	0,29	0,19	0,23
24	4	4	3	219	213	203	0,18	0,19	0,15	0,17
25	2	3	2	139	142	144	0,14	0,21	0,14	0,16
26	3	3	3	148	150	155	0,20	0,20	0,19	0,20
27	3	3	3	163	169	176	0,18	0,18	0,17	0,18
28	3	4	4	186	192	196	0,16	0,21	0,20	0,19
29	3	3	4	200	198	196	0,15	0,15	0,20	0,17
30	2	2	2	190	185	183	0,11	0,11	0,11	0,11
31	3	3	3	169	171	173	0,18	0,18	0,17	0,18
32	3	3	3	164	165	165	0,18	0,18	0,18	0,18
33	3	3	3	167	166	165	0,18	0,18	0,18	0,18
34	3	3	4	169	168	168	0,18	0,18	0,24	0,20
35	3	2	2	171	172	172	0,18	0,12	0,12	0,14
36	2	2	2	171	170	170	0,12	0,12	0,12	0,12

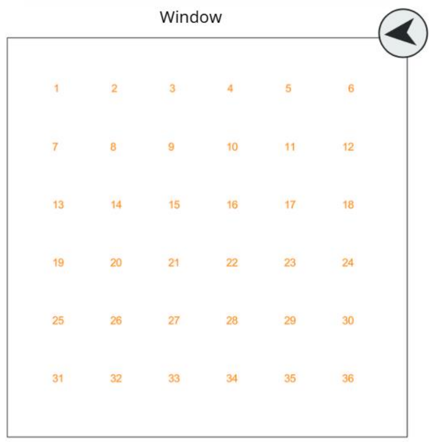

Figure 0.1 - number of the points in grid for measurement